Contents

Financial Compass **1**
Project Brief . . . 1
1. Problem Statement . . . 2
2. Proposed Solution . . . 2
3. Target Users . . . 3
4. Product Boundaries . . . 3
5. Core Modules . . . 4
6. Recommended MVP . . . 9
7. Recommended Technology Stack . . . 9
8. System Architecture . . . 10
9. Data Model . . . 10
10. AI and Normal Software Responsibilities . . . 11
11. AI Pipelines . . . 12
12. Explainable Scoring . . . 13
13. Security and Privacy Requirements . . . 13
14. Evaluation Metrics . . . 14
15. Development Roadmap . . . 14
16. Suggested Six-Week MVP Plan . . . 15
17. Demo Scenario . . . 15
18. Sample API Design . . . 16
19. Final Project Description . . . 17
20. Final Presentation Conclusion . . . 17

Financial Compass

An Explainable, Consent-Based Financial Intelligence Platform for Citizens and MSMEs

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Project Brief

Design and develop a full-stack AI/ML platform called **Financial Compass** that helps individuals and small businesses understand their financial condition using their own consented financial data.

The platform should not act as a bank, lender, credit bureau, financial advisor, tax consultant, or government authority. It should function as an **educational and decision-support system** that converts scattered financial information into understandable insights, simulations, risk indicators, scheme recommendations, fraud warnings, and compliance tasks.

The prototype should use synthetic data, manually entered data, and uploaded CSV/PDF files. Direct bank, credit bureau, GST, insurance, or government integrations should be treated as future extensions unless authorised APIs and proper compliance are available.

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1. Problem Statement

A citizen's or small business owner's financial data is usually scattered across multiple sources:

- Bank accounts
- Credit reports
- Loan statements
- Insurance policies
- Investment platforms
- Tax systems
- Government scheme portals
- GST and business records
- Bills, invoices, and payment histories

The information is available, but it is not connected in a way that ordinary users can easily understand.

As a result:

- Individuals may not know their real financial health.
- Borrowers may not understand factors affecting their credit readiness.
- Eligible citizens may miss government schemes.
- Users may fall victim to fraud and phishing.
- MSMEs may discover cash-flow or compliance problems too late.
- Policymakers may lack useful aggregated insights into financial inclusion.

The core problem is:

> People and businesses have financial data, but they lack a unified, understandable, and actionable view of that data.

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2. Proposed Solution

Financial Compass should provide a unified platform with the following capabilities:

- Financial health analysis
- Credit-readiness simulation
- Government scheme discovery
- Fraud and scam analysis
- MSME financial intelligence
- Compliance assistance
- Explainable reports
- Consent and privacy management
- Optional anonymised analytics

The system should answer questions such as:

- How healthy is my current financial position?

- How much debt am I carrying compared with my income?
- How prepared am I for emergencies?
- How would reducing an EMI affect my borrowing capacity?
- Which government schemes may be relevant to me?
- Is this message, URL, or investment offer suspicious?
- Is my business cash flow improving or deteriorating?
- Which compliance tasks are approaching?
- Which documents are missing before applying for funding?

The platform should not make final financial, legal, lending, or eligibility decisions. It should provide transparent analysis and clearly communicate uncertainty.

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3. Target Users

Individual Citizens

People who want to understand: - Income and spending - Savings - Debt - Emergency preparedness - Insurance coverage - Investment distribution - Loan readiness - Relevant government schemes - Fraud risks

Students and Young Professionals

Users who may need: - Financial literacy - Budget analysis - Education schemes - Loan simulations - Scam detection - Savings planning

MSME Owners

Small businesses that want to analyse: - Revenue - Expenses - Cash flow - Receivables - Debt - Funding readiness - Invoice data - Compliance deadlines - Financial risks

Administrators or Authorised Analysts

They may manage: - Scheme information - Compliance documents - Knowledge-base content - Anonymised aggregate dashboards - User feedback - Fraud patterns

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4. Product Boundaries

Financial Compass must clearly state that it is:

- Not an official credit-score provider
- Not a loan approval engine
- Not a replacement for a bank or lender
- Not a tax or legal advisor
- Not an official government eligibility authority
- Not a guarantee that a message is safe or fraudulent
- Not a substitute for a qualified financial professional

Use phrases such as: - “Estimated financial health indicator” - “Credit-readiness estimate” - “Potentially relevant scheme” - “Likely risk” - “Needs official verification” - “Based on the information provided”

Avoid phrases such as: - “Your loan will be approved” - “You are definitely eligible” - “This message is guaranteed to be safe” - “This is official financial advice”

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5. Core Modules

Module 1: Financial Health Analyzer

Objective Analyse a user’s income, expenses, savings, investments, insurance, and debt to produce an understandable financial-health report.

Input methods The user should be able to: - Upload a sample bank-transaction CSV - Enter monthly income - Add recurring expenses - Add loans and EMIs - Add savings - Add investments - Add insurance policies - Enter financial information manually

The prototype should not require live bank access.

Data analysis The system should calculate: - Monthly income - Monthly expenses - Essential expenses - Discretionary expenses - Monthly surplus or deficit - Savings ratio - Debt-to-income ratio - EMI burden - Emergency-fund coverage - Investment allocation - Insurance coverage status - Spending-category trends

Example formulas

Savings Ratio:

```
Savings Ratio = (Monthly Income − Monthly Expenses) / Monthly Income
```

Debt-to-Income Ratio:

```
DTI = Monthly Debt Payments / Monthly Income
```

Emergency-Fund Coverage:

```
Emergency Coverage = Liquid Savings / Monthly Essential Expenses
```

These calculations must be performed by backend code, not by an LLM.

Scores Generate transparent indicators such as: - Financial Health Indicator - Debt Risk Indicator - Emergency Preparedness Indicator - Savings Stability Indicator

Each indicator must include: - Score or category - Contributing factors - Positive factors - Risk factors - Suggested next steps - Data limitations

AI features Use AI for: - Transaction categorisation - Spending-pattern summaries - Plain-language explanations - Unusual-spending detection - Personalised educational suggestions - Monthly report generation

Example explanation: > Your monthly savings decreased mainly because food, transport, and subscription expenses increased compared with the previous three-month average.

The AI must not invent financial facts that are not present in the user's data.

Module 2: Credit-Readiness Simulator

Objective Help users understand how changes in income, debt, EMI, savings, and credit utilisation may affect their financial readiness.

This module must be a **scenario simulator**, not a loan-approval predictor.

Input variables - Monthly income - Existing EMIs - Existing debt - Proposed loan amount - Interest rate - Loan tenure - Credit utilisation - Missed-payment history, if manually provided - Monthly expenses - Savings - Business revenue, for MSME users

Outputs - Estimated EMI - Total repayment - Total interest - Debt-to-income ratio - Remaining monthly cash flow - Emergency-fund impact - Scenario risk indicators - Comparison between scenarios

Example scenario

```
Current situation:
  Monthly income: ₹60,000
  Existing EMI: ₹18,000
  DTI: 30%

After adding a new loan:
  Total EMI: ₹28,000
  DTI: 46.7%
  Risk indicator: Increased repayment pressure

After reducing existing debt:
  Total EMI: ₹22,000
  DTI: 36.7%
  Risk indicator: Improved affordability
```

AI features The AI can explain: - Why a scenario appears riskier - How debt reduction changes affordability - The trade-off between lower EMI and longer tenure - Why a user should maintain emergency savings

The AI must not claim to know a lender's internal decision criteria.

Recommended wording

Use: *"Credit Readiness Simulator"*

Do not use: *"Loan Approval Predictor"* — unless the project has access to a representative, legally usable dataset and a validated model.

Module 3: Government Scheme Finder

Objective Help users discover government schemes that may be relevant to their profile.

User profile attributes The user may voluntarily provide: - Age - State - Occupation - Student or business status - Income range - Education level - Business type - Gender, when relevant to official criteria - Disability status, when relevant and voluntarily provided - Rural or urban classification - Relevant certificates

Knowledge base Create a structured scheme database with: - Scheme name - State or national scope - Target group - Eligibility conditions - Benefits - Required documents - Application process - Official URL - Last verified date - Source document - Department or authority

AI pipeline

```
User profile
  → Extract structured attributes
  → Filter schemes by metadata
  → Retrieve official scheme documents
  → Compare profile with conditions
  → Generate an explanation
  → Show source and verification requirements
```

Result categories - Likely relevant - Potentially relevant; verify conditions - Insufficient information - Not matched by the provided profile

The system must not claim definite eligibility because official verification may be required.

Example result

```
Scheme: Student Scholarship X

Why it may match:
  - Applicant is a student.
  - State matches the scheme region.
  - Reported income is within the stated range.

Needs verification:
  - Domicile certificate.
  - Current application period.
  - Institution approval status.
```

RAG requirements Every scheme recommendation should display: - Source name - Official link - Relevant section or page - Last verified date - Required documents - Uncertainty warning

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Module 4: Fraud Protection System

Objective Analyse potentially suspicious financial communications and explain possible scam indicators.

Input types The user may submit: - SMS text - Email text - WhatsApp message - Screenshot - URL - UPI ID - Investment offer - Call transcript - Payment request

Detection targets - Requests for OTP, PIN, or passwords - Urgent payment demands - Impersonation - Fake authority claims - Unrealistic investment returns - Suspicious links - Payment redirection - Digital-arrest language - Phishing patterns - Fake customer-support messages - Requests to install unknown applications

Hybrid detection pipeline

```
Input
  → OCR or text extraction
  → Entity extraction
  → Rule-based checks
  → ML/LLM classification
  → Risk aggregation
  → Evidence-based explanation
  → Safety recommendation
```

Output - Risk level - Scam category - Suspicious entities - Detected warning signs - Explanation - Recommended action - Complaint-preparation template - Emergency steps if money has already been sent

Important design Use a hybrid system: - Rules identify explicit red flags. - AI understands context and language. - A scoring engine combines the signals. - The user sees the evidence behind the score.

The system must not guarantee that a message is safe.

Module 5: MSME Financial Intelligence

Objective Help small businesses understand their financial health, cash flow, funding readiness, and operational risks.

Input data Allow users to upload or enter: - Sales CSV - Expense CSV - Invoice records - Customer payments - Vendor payments - Loan details - GST filing status - Recurring business costs - Payroll information - Inventory expenses

Outputs - Revenue trend - Expense trend - Net cash flow - Receivables ageing - Payables analysis - EMI burden - Cash-flow forecast - Business Health Indicator - Funding Readiness Indicator - Risk assessment - Improvement checklist

AI/ML features - Invoice field extraction - Transaction categorisation - Late-payment detection - Anomaly detection - Cash-flow forecasting - Business-report generation - Funding-document checklist - Explanation of financial trends

Forecasting Start with a simple baseline such as: - Moving average - Exponential smoothing - Linear regression

Then compare it with an ML model if sufficient sample data is available.

> Do not use deep learning unless the dataset justifies it.

Example insight > Revenue increased over the previous three months, but cash availability decreased because customer payments are arriving later and loan repayments increased.

Module 6: Compliance Copilot

Objective Help MSMEs track selected compliance tasks and deadlines.

Recommended initial scope Choose one or two areas: - GST-return reminders - Business licence renewals - Startup compliance checklists - Invoice and record requirements - MSME document readiness

> Do not attempt to cover all Indian tax, labour, company, and industry laws in the first version.

Features - Business profile - Compliance calendar - Task creation - Deadline tracking - Document upload - Missing-document detection - Reminder notifications - Assigned staff - Completion status - Audit history

AI features - Extract deadlines from uploaded notices - Classify compliance tasks - Summarise official circulars - Identify missing documents - Answer questions from approved documents - Generate a compliance checklist

Safety Each AI answer must show: - Source - Date - Applicable scope - Confidence level - Professional-review warning when necessary

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Module 7: Financial Inclusion Analytics

Objective Provide anonymised and aggregated insights for institutions, NGOs, or policymakers.

Possible analytics - Underbanked regions - Low credit-access areas - Insurance-coverage gaps - Digital-payment adoption - Common financial risks - Scheme-awareness trends - MSME funding-readiness trends

Privacy requirements The analytics module must: - Use anonymised data - Use aggregation thresholds - Prevent re-identification - Avoid exposing individual financial records - Separate user-level data from analytics data - Include explicit consent for data use

> For a college project, this module can be implemented with generated data and presented as a future extension.

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6. Recommended MVP

Do not implement every module initially.

Build the following four modules as the working prototype: 1. Financial Health Analyzer 2. Credit-Readiness Simulator 3. Government Scheme Finder 4. Fraud Protection System

Add a basic MSME dashboard if time permits.

MVP user journey

```
Register
  → Read and provide purpose-specific consent
  → Upload synthetic financial CSV
  → View financial health dashboard
  → Simulate a loan scenario
  → Search potentially relevant schemes
  → Scan a suspicious message
  → Export a financial report
```

Future extensions - MSME financial intelligence - Compliance Copilot - Voice interface - Regional-language support - Direct Account Aggregator integration - GST integration - Insurance integration - Policymaker analytics - Mobile application - Financial advisor escalation

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7. Recommended Technology Stack

Frontend - Next.js - TypeScript - Tailwind CSS - shadcn/ui - Recharts - React Hook Form - React Query

Backend - FastAPI - Python - Pydantic - SQLAlchemy - PostgreSQL

> Python is recommended because it supports the financial-analysis, document-processing, forecasting, and ML components in one ecosystem.

AI and ML - LLM API or compatible open-source model - Sentence-transformers for embeddings - pgvector for vector search - scikit-learn for classification and anomaly detection - Pandas for financial data analysis - OCR library or document OCR API - Optional speech-to-text model - Optional translation model

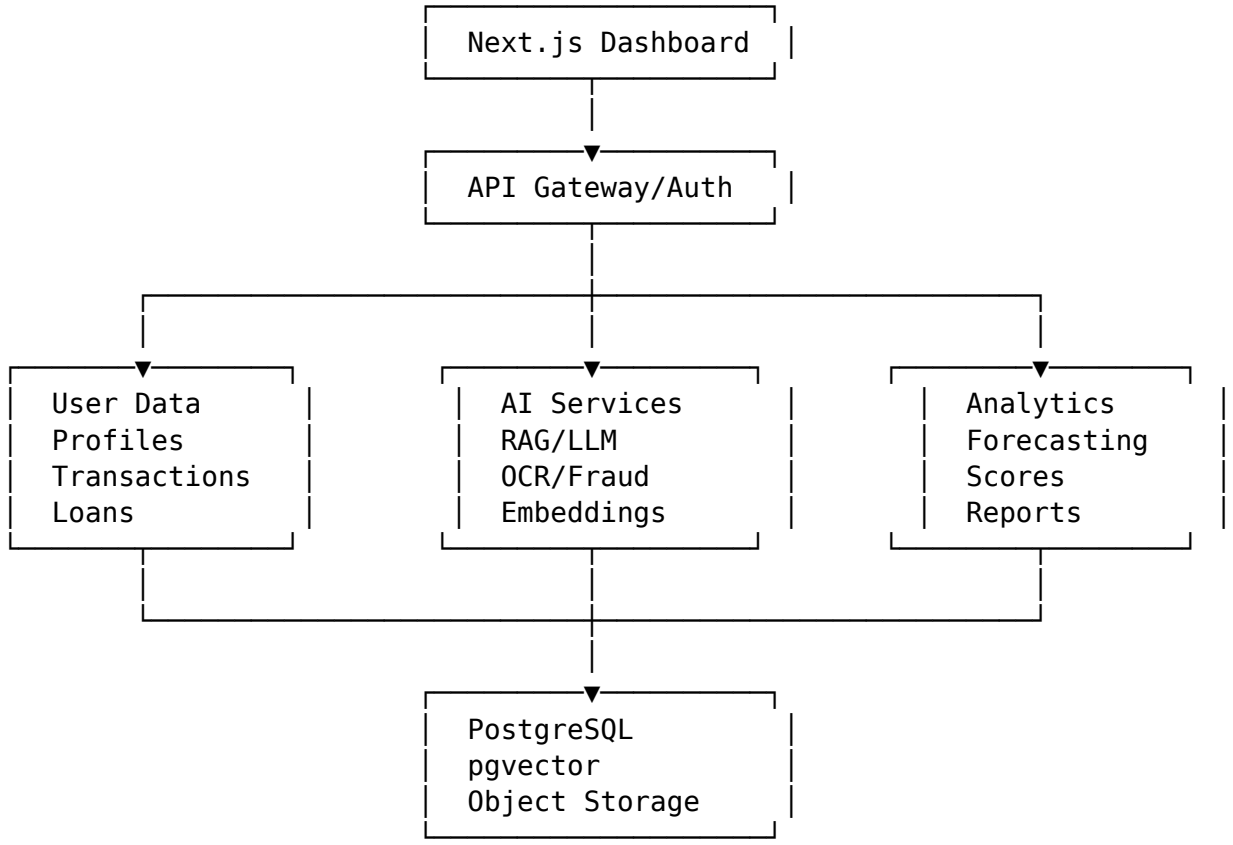
Infrastructure - Redis - Celery - S3-compatible object storage - Docker - GitHub Actions - Sentry - OpenTelemetry

Deployment - Frontend: Vercel or equivalent - Backend: Render, Railway, Fly.io, or cloud VM - Database: Managed PostgreSQL - Storage: S3-compatible object storage

> Start with a modular monolith. Do not begin with multiple microservices.

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8. System Architecture



Background processing Use asynchronous jobs for: - CSV parsing - PDF extraction - OCR - Embedding generation - Large-document processing - Forecast generation - Report generation - Email notifications

```
Upload → Background job → Process → Store → Notify frontend
```

---

9. Data Model

Use tables such as:

```
users
profiles
organisations
consents
financial_accounts
transactions
transaction_categories
loans
insurance_policies
investments
health_scores
```

```
credit_scenarios
schemes
scheme_eligibility_rules
scheme_matches
fraud_scans
businesses
invoices
cashflow_records
compliance_tasks
documents
document_versions
document_chunks
ai_predictions
ai_traces
feedback
audit_logs
```

Consent table

```
consents
---------
id
user_id
data_type
purpose
recipient
created_at
expires_at
revoked_at
status
```

The system should record exactly: - What data was shared - Why it was shared - With whom it was shared - When consent was given - When it expires - Whether it was revoked

> The Account Aggregator model in India is specifically based on consent-driven financial-data sharing. (financialservices.gov.in)

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10. AI and Normal Software Responsibilities

Feature	Normal Backend Logic	AI/ML
Income calculation	Yes	No
Expense totals	Yes	No
EMI calculation	Yes	No
Savings ratio	Yes	No
Transaction categorisation	Rules plus ML	Yes
Financial report	Data aggregation	Explanation

Feature	Normal Backend Logic	AI/ML
Scheme filtering	Metadata rules	Matching and explanation
Fraud detection	Red-flag rules	Context classification
Cash-flow forecast	Statistical/ML model	Forecasting
Compliance reminders	Yes	Deadline extraction
Document search	Database/vector search	Embeddings
Final loan approval	No	Must not be claimed

> Use AI where interpretation, extraction, classification, or explanation is needed. Use deterministic software for calculations, permissions, deadlines, and financial records.

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11. AI Pipelines

Financial transaction pipeline

```
CSV upload
  → Validate columns
  → Normalise dates and amounts
  → Remove duplicates
  → Extract transaction descriptions
  → Categorise transactions
  → Store structured records
  → Calculate metrics
  → Generate explanation
```

Scheme recommendation pipeline

```
User profile
  → Validate profile
  → Apply metadata filters
  → Retrieve scheme documents
  → Compare eligibility conditions
  → Generate matched-result explanation
  → Attach official sources
  → Mark verification requirements
```

Fraud analysis pipeline

```
Message or screenshot
  → OCR if required
  → Text normalisation
  → Entity extraction
  → Rule checks
  → ML/LLM classification
  → Risk aggregation
  → Evidence generation
  → Safety guidance
```

Document compliance pipeline

```
Upload document
  → Validate file
  → Extract text
  → Detect document type
  → Extract dates and obligations
  → Store source passages
  → Create compliance task
  → Notify business user
```

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12. Explainable Scoring

Every score must be explainable.

Example financial health output

```
Financial Health Indicator: Moderate

Positive factors:
  - Monthly income is stable.
  - Savings are present.
  - No recent missed payment is recorded.

Risk factors:
  - Debt payments consume a large part of monthly income.
  - Emergency savings cover fewer months than the selected target.
  - Discretionary spending increased recently.

Suggested actions:
  - Review recurring subscriptions.
  - Compare EMI-reduction options.
  - Build an emergency reserve gradually.
```

Do not display a score without explaining: - How it was calculated - Which data was used - What data is missing - How the user can improve it

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13. Security and Privacy Requirements

The project handles sensitive financial information, so security must be a core part of the design.

Implement: - Secure authentication - Hashed passwords - HTTPS in deployment - User-level data isolation - Role-based access control - Input validation - File-type and file-size validation - Malware scanning where possible - Encryption at rest where supported - Encryption in transit - API rate limiting - Audit logs - Secure secret man-

agement - PII masking in logs - Data deletion - Consent revocation - Session expiration - Prompt-injection protection for uploaded documents

Never log: - Full account numbers - Passwords - OTPs - UPI PINs - Authentication tokens - Unmasked identity documents

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14. Evaluation Metrics

Transaction categorisation - Accuracy - Precision - Recall - Macro F1 score - Uncategorised transaction percentage

Scheme recommendation - Recommendation precision - Recommendation recall - Citation accuracy - Eligibility-condition extraction accuracy - Verification-warning accuracy

Fraud detection - Precision - Recall - False-positive rate - Scam-category accuracy - Explanation quality

Forecasting - Mean Absolute Error - Root Mean Squared Error - Forecast error by month - Performance against a baseline

RAG system - Retrieval recall - Context precision - Answer faithfulness - Citation correctness - Abstention accuracy - Hallucination rate

Product metrics - Dashboard loading time - AI response latency - Cost per analysis - Report-generation time - User feedback score - Number of successfully completed actions

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15. Development Roadmap

Phase 1: Foundation - Create repository - Set up frontend and backend - Configure PostgreSQL - Add authentication - Add user profile - Create dashboard layout - Add database migrations

Phase 2: Financial data - Create CSV upload - Validate and normalise transactions - Add categorisation - Store transaction records - Create spending charts - Calculate financial indicators

Phase 3: Credit simulator - Add loan form - Calculate EMI and repayment - Add scenario comparison - Add DTI analysis - Generate AI explanations - Store simulation history

Phase 4: Scheme finder - Create scheme database - Add official documents - Build metadata filters - Add embeddings and vector search - Generate cited recommendations - Add verification warnings

Phase 5: Fraud detection - Add message input - Add screenshot OCR - Add rule-based checks - Add AI classification - Add risk explanations - Add safety guidance

Phase 6: MSME extension - Add business profile - Add invoice upload - Add revenue and expense tracking - Add cash-flow reports - Add basic forecast - Add funding-readiness checklist

Phase 7: Quality and deployment - Create evaluation datasets - Test incorrect inputs - Add permission tests - Add rate limits - Add logging and monitoring - Deploy frontend and backend - Write documentation - Record a product demo

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16. Suggested Six-Week MVP Plan

Week	Deliverable
1	Authentication, profile, database, dashboard
2	CSV upload, transaction normalisation, categorisation
3	Financial-health calculations and visualisations
4	Credit-readiness simulator and scenario comparison
5	Scheme finder and fraud-analysis module
6	Testing, evaluation, security, deployment, presentation

If working as a team, divide responsibilities as follows:

Frontend developer - Dashboard - Forms - Charts - Reports - Responsive interface

Backend developer - APIs - Database - Authentication - File processing - Permission system

AI/ML developer - Categorisation - RAG - Fraud classification - Forecasting - Evaluation

Integration and documentation developer - Background jobs - Deployment - Monitoring - API documentation - Demo data - Testing

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17. Demo Scenario

Use synthetic data for the final demonstration.

Citizen demonstration 1. User registers. 2. User gives consent for analysis. 3. User uploads a sample transaction CSV. 4. Platform categorises transactions. 5. Dashboard displays income, expenses, savings, and debt. 6. Financial Health Indicator is generated. 7. User enters a new loan scenario. 8. Platform shows EMI, DTI, and cash-flow impact. 9. User enters profile details. 10. Platform recommends potentially relevant government schemes. 11. User uploads a suspicious SMS screenshot. 12. Platform highlights scam indicators. 13. User downloads the final report.

MSME demonstration 1. Business owner creates a business profile. 2. Uploads sales and expense CSV. 3. Uploads sample invoices. 4. Platform extracts invoice

fields. 5. Dashboard shows revenue and cash-flow trends. 6. Platform detects delayed receivables. 7. Cash-flow forecast is generated. 8. Funding-readiness checklist is displayed. 9. A compliance deadline is extracted from a document. 10. Reminder task is created.

---

18. Sample API Design

Authentication

```
POST /api/auth/register
POST /api/auth/login
POST /api/auth/logout
```

Financial data

```
POST /api/transactions/upload
GET  /api/transactions
GET  /api/financial-health
GET  /api/spending-summary
```

Credit simulation

```
POST /api/credit-scenarios
GET  /api/credit-scenarios
POST /api/credit-scenarios/compare
```

Scheme finder

```
GET  /api/schemes
POST /api/schemes/match
GET  /api/schemes/{id}
```

Fraud analysis

```
POST /api/fraud/analyse-text
POST /api/fraud/analyse-image
GET  /api/fraud/scans
```

MSME

```
POST /api/businesses
POST /api/businesses/transactions/upload
POST /api/businesses/invoices/upload
GET  /api/businesses/cashflow
GET  /api/businesses/funding-readiness
```

Compliance

```
POST  /api/compliance/documents
GET   /api/compliance/tasks
PATCH /api/compliance/tasks/{id}
```

---

19. Final Project Description

Financial Compass is an explainable, consent-based financial intelligence platform designed for citizens and MSMEs. It consolidates user-provided financial information, analyses income, expenses, debt, savings, and cash flow, and converts the results into understandable financial-health indicators, credit-readiness simulations, government-scheme recommendations, fraud warnings, and compliance tasks. The platform combines full-stack development, data analytics, machine learning, retrieval-augmented generation, document processing, and secure role-based workflows. It does not replace banks, credit bureaus, financial advisors, tax professionals, or government authorities; instead, it provides educational and decision-support insights based on available data.

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20. Final Presentation Conclusion

Financial Compass is not merely a financial-management application. It is an explainable financial-intelligence platform that converts scattered, user-consented data into practical insights and action plans.

Individuals can understand their financial health and credit readiness. MSMEs can identify cash-flow and compliance risks before they become serious problems. Users can discover potentially relevant government schemes and receive warnings about suspicious financial communications.

The vision of Financial Compass is simple:

Turn financial data into understandable direction, so citizens and businesses can make smarter and safer financial decisions.